Identifying Radioactive Calibration Sources in Single Bars of the X17 Detector with Gradient-Boosted Trees

Samy Rahmani

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Abstract

Radioactive sources are used to check and calibrate the X17 detector, and the lab keeps a registry of which source sits where. Keeping that registry correct today means asking people. We present a way to read the source identity directly from the data of a single scintillator bar. We simulate the response of the detector to four lab sources (Cs-137, Co-60, Bi-207 and Na-22) and to cosmic muons, build a large set of labelled data windows that cover different durations, source strengths, bars, source positions, energy resolutions and energy scales, and train an XGBoost classifier on the energy spectrum of each window. A template fit based on a Poisson likelihood serves as a baseline. On 20 000 test windows the two methods reach similar accuracy (66.2% and 66.7%). The accuracy is set by the number of source events in the window: it is above 90% from a few hundred source events and close to 100% from a few thousand, and it depends only weakly on the energy resolution and the energy scale in the ranges we studied. At the real activities of the lab sources, Cs-137 and Bi-207 are identified in every bar, Co-60 in most bars, and Na-22 only with long windows. A source that was not part of the training can be recognised as unknown only when the window holds a large number of source events. The method is ready to be compared with measurements taken in the lab.

1 Introduction

The X17 experiment looks for a hypothetical light boson through the electron–positron pairs it would produce in nuclear transitions [1]. Its detector is built from plastic scintillator bars, and the response of these bars has to be checked with radioactive sources. The lab uses several sealed sources for this, and they are placed at different positions along the bars.

A registry of these sources, with their position and the time they were used, is needed to interpret the data afterwards. At present the registry depends on people writing down what they did. We want to be able to confirm from the data alone which source was in front of a bar, without asking anyone. This is a classification problem: given the hits recorded by one bar during a time window, decide which source, if any, was present.

The problem is harder than it looks. The lab sources are old, and some of them are now weak, so a window may contain only a handful of source events next to a continuous background of cosmic muons. The energy resolution of a bar and the calibration of its energy scale are known only approximately and change from bar to bar and over time. The four sources we use, Cs-137, Co-60, Bi-207 and Na-22, have gamma lines between 0.5 and 1.8 MeV, and in a plastic scintillator these lines show up only as Compton continua with broad edges, not as photopeaks. Separating them therefore relies on the shape of a smooth spectrum.

Approach

We split the work into three steps.

1. **Simulation.** We simulate the detector with Geant4 [2] for the four lab sources (Cs-137, Co-60, Bi-207, Na-22, lab sources 312, 319, 308 and 332) and for cosmic muons, with the source capsule placed at many positions along the bars. A parametrised response (energy resolution, energy scale, threshold, position variable) is applied on top of the simulation.

2. **Training.** From the simulated events we build a large number of labelled time windows and train an XGBoost classifier [4] on their energy spectra. We compare it with a classical template fit.

3. **Validation on lab data.** The final test is to run the trained model on data taken with the real sources in the lab. This step has not been done yet. This report prepares it: it fixes the method, measures how the performance depends on the conditions, and states what is needed from the lab data.

XGBoost and why it fits this problem

XGBoost [4] is an implementation of gradient-boosted decision trees [5]. A decision tree classifies an example by asking a sequence of yes/no questions about its input values (for example, "is the fraction of events between 0.4 and 0.5 MeV larger than 3%?"). A single tree is weak. Boosting builds many small trees one after the other, and each new tree is trained to correct the mistakes of the previous ones. The final score of each class is the sum of the outputs of all trees, turned into probabilities with a softmax function. Training stops when the loss on a separate validation set no longer improves.

Three properties make this method a good fit here. First, our input is a fixed-length table (the normalised energy histogram and a few summary numbers), which is the kind of data tree ensembles handle best, and no hand-made preprocessing is needed. Second, the model outputs class probabilities, which we use to judge how confident an identification is and to flag unknown sources. Third, training is fast, and the trained model has a feature importance that we can compare with the physics (the gain is concentrated at the Compton edges, see Section 3).

The template fit plays the role of a physics-based reference: if a flexible model cannot beat a fit that uses the known spectra, it is not adding anything. We report both.

2 Methodology

The pipeline has five stages: simulation, a response model applied to the simulated hits, construction of labelled windows, classification, and evaluation. Every stage is driven by configuration files, with one random seed for the whole chain, so that a run can be reproduced exactly.

2.1 Simulation

The detector geometry is a ring of 16 trapezoidal polystyrene bars, each 140 cm long and 9.5 cm thick, with a width going from about 10.1 to 6.3 cm. The simulation uses Geant4 [2] with the FTFP_BERT physics list and the standard electromagnetic option 4 together with radioactive decay for the sources (option 3 for muons). Hits are grouped per bar and per event: all energy deposits of one bar that follow each other within 10 ns form a single hit, with the summed deposited energy and the energy-weighted mean position along the bar.

Sources. We simulate the four lab sources listed in Table 1. Each source is a generic encapsulated disk (5 cm diameter, 1 cm thick, plexiglass body, small polyethylene active deposit) and not a model of the real capsule. The capsule sits at $x = 0$, $y = -30$ cm and is moved along the bar axis, with its position drawn uniformly between -70 and $+70$ cm. For each source we simulate 20 000 000 primary decays. The activity of each source at the date of the simulation follows from its reference activity, its reference date and the half-life. Because the sources are old, the present activities differ strongly: Cs-137 and Bi-207 are still strong, while Co-60 is of order 1.2×10^2 Bq and Na-22 of order 8.8×10^0 Bq.

isotope	lab id	reference activity (Bq)	reference date	activity today (Bq)	primaries
Bi-207	308	3.77×10^5	1977-09-02	1.28×10^5	20 000 000
Cs-137	312	3.70×10^6	1972-10-01	1.07×10^6	20 000 000
Co-60	319	3.18×10^5	1966-12-13	1.22×10^2	20 000 000
Na-22	332	2.78×10^5	1987-11-23	8.80×10^0	20 000 000

Table 1: Lab sources used in the simulation.

Cosmic muons. Muons are generated on a 2×2 m^2 plane above the detector with the parametrisation of Guan et al. [3]. The total rate through the plane is 510 Hz, and 20 000 000 generated muons correspond to 39 250 s of real time. The rate seen by each bar follows from the fraction of muons that deposit energy in that bar.

Time structure. Each hit carries the time since the decay of the primary ion. For Cs-137 the gamma ray of the daughter Ba-137m is emitted about 221 s (mean life) after the decay.

Sanity check of the simulated spectra. Figure 1 shows the deposited-energy spectra of the four sources and of the muons before any response is applied. The steepest fall near each expected Compton edge sits at the nominal position, as Table 2 shows (column "no smearing"). With the detector resolution applied, the position of the steepest fall moves to higher energy, which is the expected effect of smearing a falling edge.

source	expected (MeV)	no smearing	$R = 0.05$	$R = 0.1$	$R = 0.2$
Cs-137	0.477	0.48	0.50	0.52	0.52
Co-60	0.963	0.96	1.06	1.00	1.00
Co-60	1.118	1.12	1.12	1.16	1.07
Bi-207	0.393	0.39	0.41	0.42	0.43
Bi-207	0.858	0.86	0.88	0.92	0.88
Na-22	0.341	0.34	0.35	0.39	0.39
Na-22	1.062	1.06	1.11	1.13	1.12

Table 2: Position (MeV) of the steepest fall near each expected Compton edge, in the simulated spectra without smearing and with smearing for three resolutions R (σ/E at 0.662 MeV).

2.2 Detector response

The simulation gives the true deposited energy. The response is applied afterwards, so that the resolution and the calibration can be changed without a new simulation.

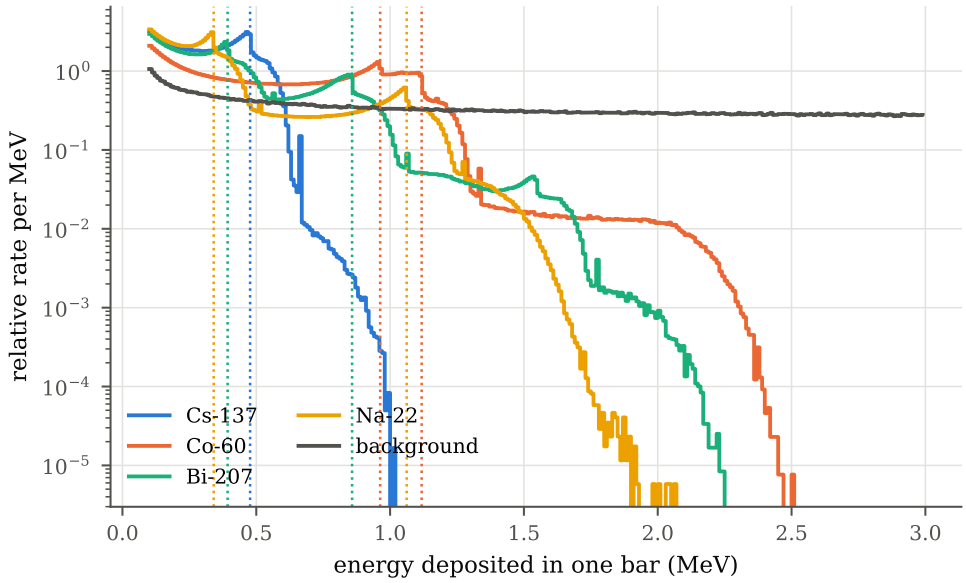


Figure 1: Energy deposited in one bar by single events, before the detector response. Dotted lines mark the expected Compton edges of each source. Muons give a flat continuum that extends to much higher energies.

Energy. The true energy E is smeared with a Gaussian of width

$$\sigma(E) = k\,\sqrt{a^2 E + b^2 E^2 + c^2}, \tag{1}$$

where a is the stochastic term, b the constant term and c the noise term, with the values of Table 3. The factor k rescales the whole curve so that the relative resolution at 0.662 MeV equals the resolution R chosen for the window. The energy is then multiplied by an energy scale s, and events below the analysis threshold (0.1 MeV plus a per-window shift) are removed.

Position. Each hit gets a position variable $x = 2z/\lambda + \epsilon$, where z is the true position along the bar, λ the attenuation length and ϵ a Gaussian noise term.

Parameter values. The values of a, b, c, λ and the position noise are plausible values chosen for this study. They have not been measured on the real bars. For this reason R and s are not fixed: they are nuisance parameters, drawn independently for each window over wide ranges (Table 3). A classifier that works over these ranges does not depend on the exact response of one bar.

quantity	value
stochastic term a	0.1
constant term b	0.02
noise term c (MeV)	0.01
attenuation length (cm)	100
position noise (cm)	0.1
resolution at 0.662 MeV	uniform [0.03, 0.25]
energy scale	uniform [0.9, 1.1]
threshold shift (MeV)	uniform [0, 0.05]

Table 3: Parameters of the response model and ranges of the nuisance parameters drawn for each window.

2.3 Windows

The classifier works on time windows, not on single events: a window is the set of hits in one bar during a given time. This is also how the data will arrive in the lab.

Window content. A window is defined by a bar, a duration, a source (or none) and a number of source events. The number of muon events is Poisson distributed with the muon rate of the bar times the duration. The events are sampled with replacement from the simulated events of that bar. We build two kinds of windows:

- *Training-type windows*: the number of source events is drawn log-uniformly between 1 and 20 000, and the duration log-uniformly between 0.1 and 3600 s, independently. This covers the whole range from almost empty to source-dominated windows. Figure 2 shows the distributions.
- *Realistic windows*: the number of source events is Poisson distributed with the real source activity, times the efficiency of the bar, times the duration (1 to 3600 s). They show what the detector would see with the lab sources as they are today.

The two kinds are drawn from the same simulated events and the same train/test split. Windows with more than 100 000 events are subsampled to that number, keeping the muon-to-source ratio.

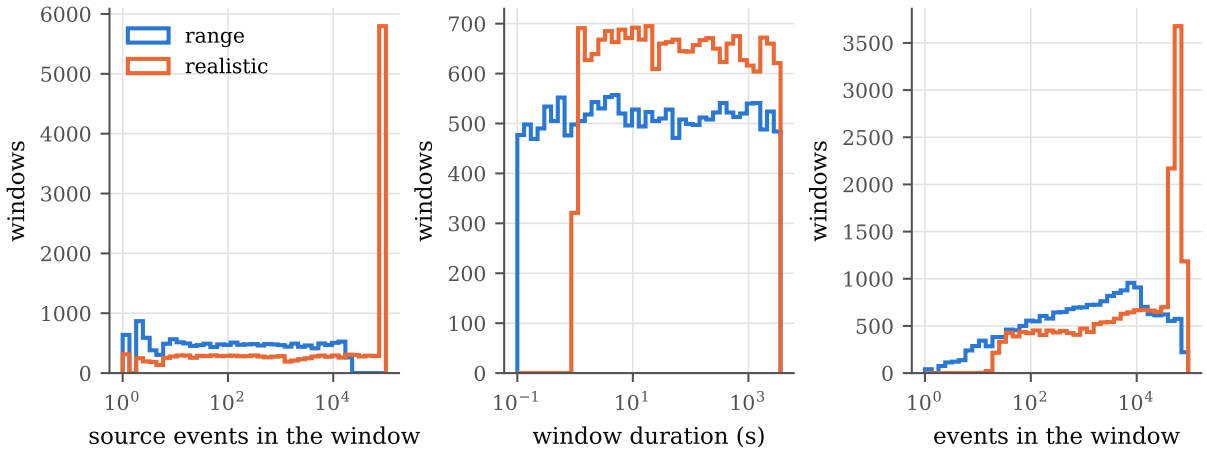


Figure 2: Distributions of the source events, the duration and the total number of events in the two kinds of test windows. The peak at 100 000 events in the realistic windows comes from the cap on the window size.

Classes and balance. There are five classes: the four sources and "background", which contains only muons. Each split holds the same number of windows for every class (Table 4).

Splits without leakage. A window never shares simulated events with a window of another split. For the sources, the capsule positions are grouped in 5 cm bins and each bin belongs to one split (70% train, 15% validation, 15% test), so that the test windows come from capsule positions the model has not seen. For the muons the events are split in blocks of 20 000 consecutive events. The code checks the disjointness of the splits when the pools are built.

Reproducibility. Two runs with the same seed produce files that are identical byte for byte.

class	train	validation	test
Cs-137	8 000	1 000	4 000
Co-60	8 000	1 000	4 000
Bi-207	8 000	1 000	4 000
Na-22	8 000	1 000	4 000
background	8 000	1 000	4 000

Table 4: Number of training-type windows per class and split. The realistic test set has 20 000 windows with the same class balance.

2.4 Features

For each window we compute the normalised energy histogram (102 input columns in total, of which 95 are histogram bins). The bins are 0.025 MeV wide up to 1.5 MeV, then widen up to 1 GeV, so that muons with large deposits are kept in the last bins. The histogram is normalised to unit sum, which removes the total count. We add the number of events in the window and six summary statistics of the position variable (standard deviation, inter-quartile range, median absolute deviation, 90–10 percentile range, skewness and excess kurtosis).

The model sees only these feature columns. Everything that describes the window (label, source, true counts, duration, the nuisance values) is kept in separate columns that the model code refuses to read.

2.5 XGBoost classifier

We use the multiclass softmax objective of XGBoost with histogram-based trees, a maximum depth of 6, a learning rate of 0.1, row subsampling of 0.8, column subsampling of 0.8 and early stopping on the validation log-loss (patience of 30 rounds). The best round was 180. The validation set is used only to stop training. Windows of the test split are used only for the final numbers. The same procedure, with the class set restricted, gives the model used for the unseen-source test (Section 3.9).

2.6 Template fit

The baseline needs no training in the usual sense. For each source we build a template: the mean normalised histogram of the training windows of that source with a source fraction of at least 0.95. The same is done for the muon background. For a window with histogram counts n_i and total N, we model the expected counts as

$$\mu_i = N\,[(1 - f)\,b_i + f\,t_i]\,, \tag{2}$$

where b_i is the background template, t_i the source template and f the source fraction, fitted in $[0, 1]$ by maximising the Poisson likelihood. The window is assigned to the source with the highest likelihood, unless the improvement over the background-only model, $2\Delta \ln L$, is smaller than a threshold, in which case it is called background. The threshold is chosen on the validation split from a grid. The goodness of fit is the Poisson deviance divided by the number of degrees of freedom.

2.7 Evaluation

All numbers come from windows of the test split. We report:

- the accuracy, confusion matrices and per-class precision and recall;
- the accuracy as a function of the window properties: events, source events, source fraction, resolution, energy scale, source position and bar;
- the rate at which Co-60 and Bi-207 are swapped, as a function of resolution, energy scale and number of source events. These two sources have the closest gamma energies of the set;
- the calibration of the XGBoost probabilities (reliability diagram, expected calibration error, Brier score);
- the recall for each source and each bar on the realistic windows;
- the detection of an unseen source: we train a second pair of models without one source and ask whether windows of that source can be told apart from the known classes, using the score $1 - p_{\max}$ (XGBoost) and the goodness of fit (template fit). We report ROC curves and the true positive rate at fixed false positive rates. No threshold is fixed in this work.

3 Results

3.1 What the classifier sees

Figure 3 shows the mean energy spectrum of source-dominated windows with many source events. The four sources have clearly different shapes, and the muon background is nearly flat above the Compton edges. Resolution blurs the edges (Figure 4), but the main differences between sources remain visible over the whole range $R = 0.03$ to 0.25. With few source events the picture is different: a window with 30 to 60 source events (Figure 5) is a sparse histogram, and the identification has to rely on a few tens of entries.

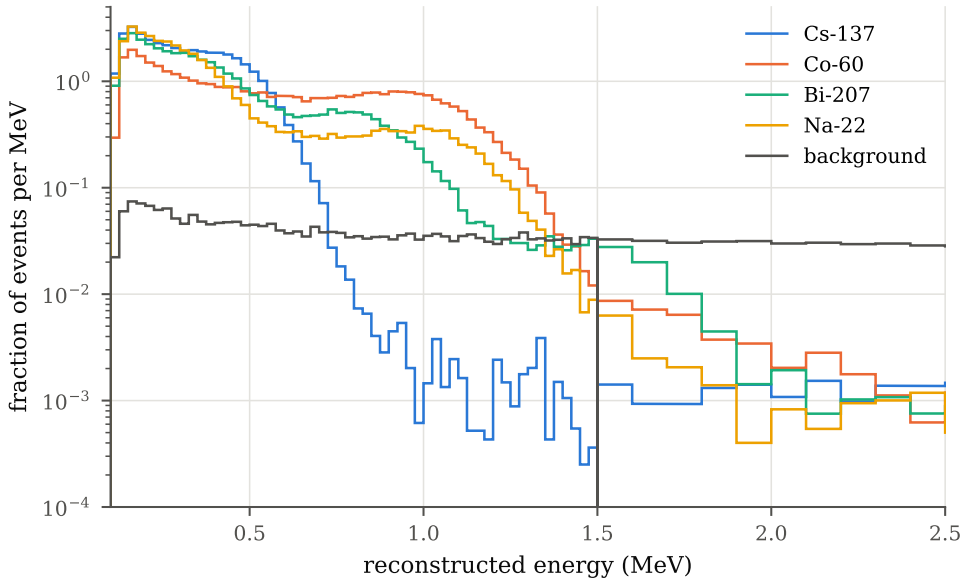


Figure 3: Mean reconstructed spectrum of source-dominated windows (at least 1000 source events, source fraction of at least 0.9) at a resolution between 0.10 and 0.14 and an energy scale close to 1.

3.2 Training and what the model uses

The validation loss of the XGBoost model flattens after about 180 rounds (Figure 6), and the validation accuracy at that point is 66.1%. The gain of the trees is concentrated in the energy

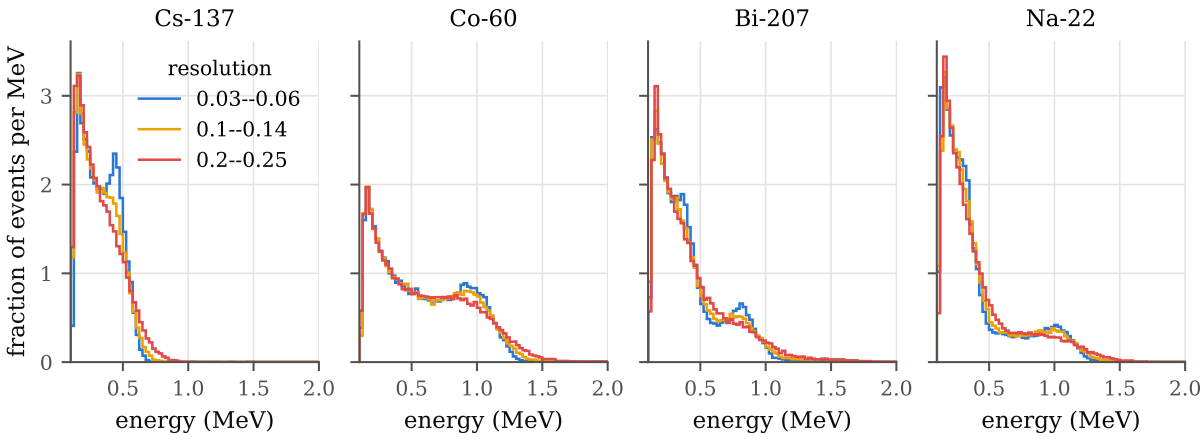


Figure 4: The same spectra for three resolution ranges (σ/E at 0.662 MeV).

histogram (97.2% of the total), and within the histogram it peaks at the energies of the Compton edges of the four sources (Figure 7). The event count (0.7%) and the position shape (2.1%) contribute little. The model has therefore learned what the physics suggests: the identity is carried by the spectral shape around the edges.

3.3 Overall performance

On the 20 000 training-type test windows the accuracy is 66.2% for XGBoost and 66.7% for the template fit. On the 20 000 realistic windows it is 78.4% and 78.0%. These numbers mix windows of very different quality, since the training-type windows include windows with a single source event. The confusion matrices (Figure 8) show where the errors come from: on the training-type windows about one fifth of the source windows are called background, which is what happens when the source events are too few to stand out of the muon spectrum. Tables 5 and 6 give precision and recall per class.

class	support	XGBoost P	XGBoost R	template P	template R
Cs-137	4 000	0.760	0.667	0.816	0.665
Co-60	4 000	0.796	0.654	0.725	0.673
Bi-207	4 000	0.694	0.553	0.753	0.537
Na-22	4 000	0.726	0.575	0.746	0.595
background	4 000	0.504	0.864	0.495	0.864

Table 5: Precision (P) and recall (R) per class on the training-type test windows.

class	support	XGBoost P	XGBoost R	template P	template R
Cs-137	4 000	0.830	1.000	0.882	0.999
Co-60	4 000	0.872	0.793	0.824	0.800
Bi-207	4 000	0.812	1.000	0.861	0.977
Na-22	4 000	0.760	0.286	0.779	0.290
background	4 000	0.659	0.842	0.599	0.832

Table 6: Precision (P) and recall (R) per class on the realistic test windows.

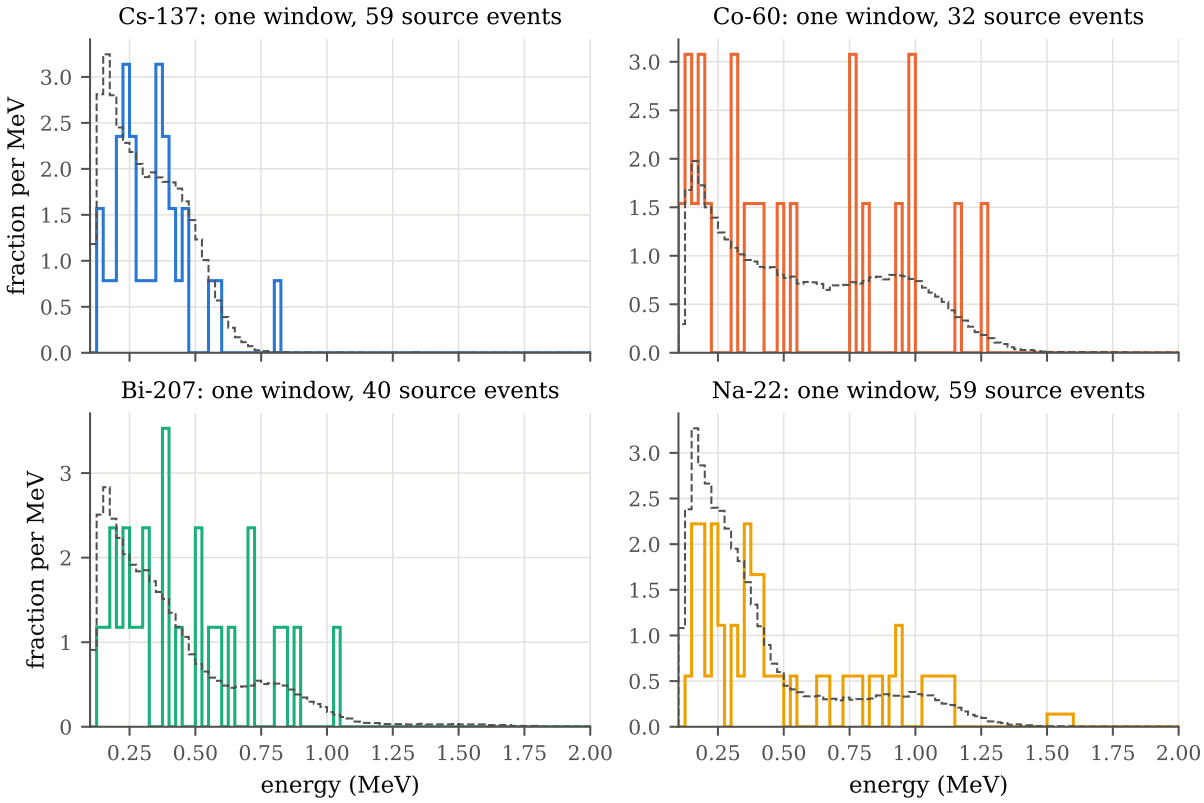


Figure 5: Single windows with 30 to 60 source events (solid) compared with the mean spectrum of source-dominated windows (dashed).

3.4 Dependence on the number of source events

The number of source events, not the total number of events, decides the outcome. Figure 9 shows the accuracy against three quantities. The accuracy against the total number of events is flat or even decreases at high counts, because a large window is often dominated by muons. The accuracy rises steadily with the source fraction and with the number of source events. Table 7 gives the numbers for windows in which the source makes up at least half of the events.

source events	windows	XGBoost	template fit
1–3	32	0.094	0.188
3–10	188	0.330	0.463
10–30	367	0.559	0.619
30–100	585	0.759	0.814
100–300	773	0.904	0.915
300–1000	1 040	0.977	0.973
1000–3000	1 060	0.996	0.986
$\geq$ 3000	2 371	1.000	0.995

Table 7: Accuracy against the number of source events, for windows with a source fraction of at least 0.5.

The two-dimensional view (Figure 10) shows the same behaviour. A window is easy when it has many events and a high source fraction. A source fraction below 0.1 gives poor identification at any size of window that we simulated, because the source spectrum is a small correction on the muon spectrum.

The recall of each class (Figure 11) rises with the number of source events for every source, and

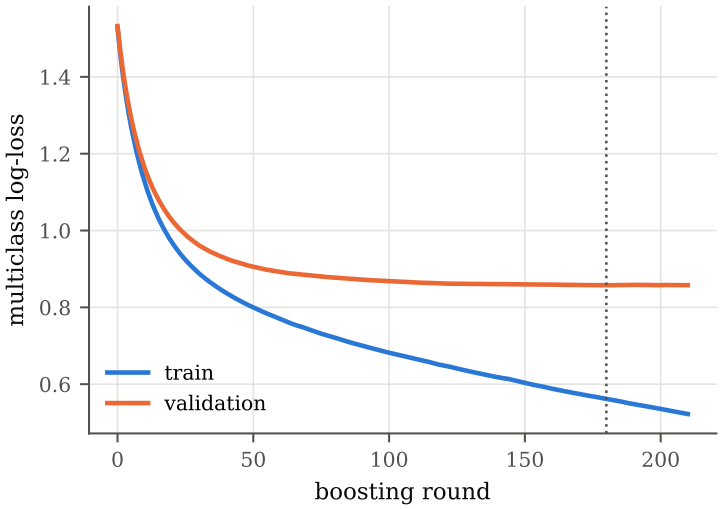


Figure 6: Training and validation log-loss of the XGBoost model. The dotted line marks the round selected by early stopping.

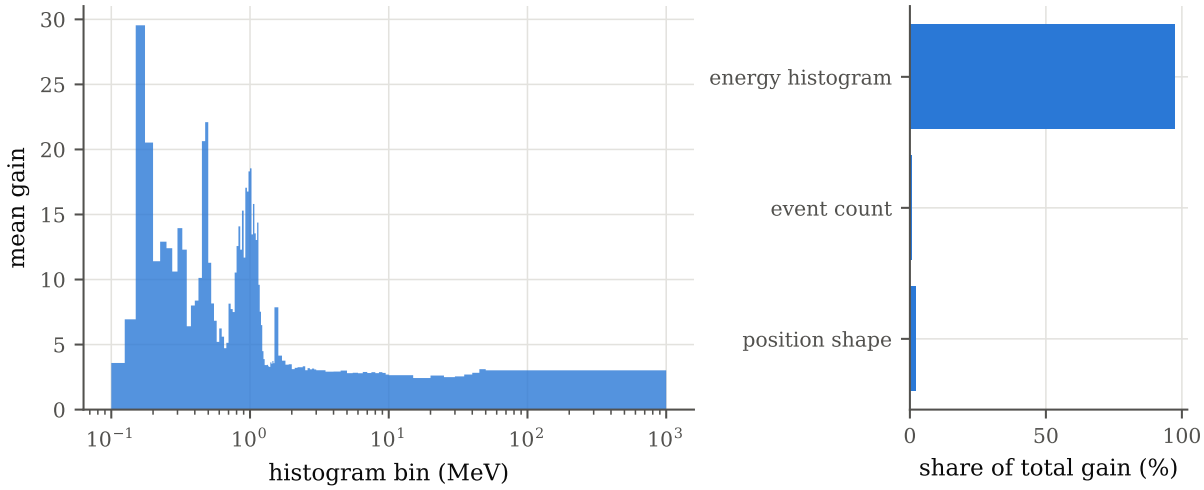


Figure 7: Left: mean gain of the histogram bins of the XGBoost model. Right: share of the total gain by feature group.

the sources do not reach the same recall at the same count. Cs-137 is the easiest, since its single edge is far from the others. Bi-207 and Na-22 need the most events.

3.5 Energy resolution, energy scale and geometry

Figure 12 shows the accuracy against the two response nuisance parameters, for windows with a source fraction of at least 0.5. When the window has at least 100 source events, the accuracy is high over the whole range of resolutions (0.03 to 0.25) and energy scales (0.9 to 1.1). At 10 to 100 source events the statistics dominate, and the dependence on the nuisance parameters is small compared with the dependence on the number of events; the worst resolution and the edges of the scale range are the least favourable. In practice, a calibration of the energy scale to within a few percent and a resolution known to within the range studied are enough for windows that contain a few hundred source events.

Figure 13 shows the accuracy against the position of the capsule along the bar and against the bar number. There is no strong dependence on the bar. The dependence on the capsule position is mild; the test windows come from capsule positions that the model never saw in training.

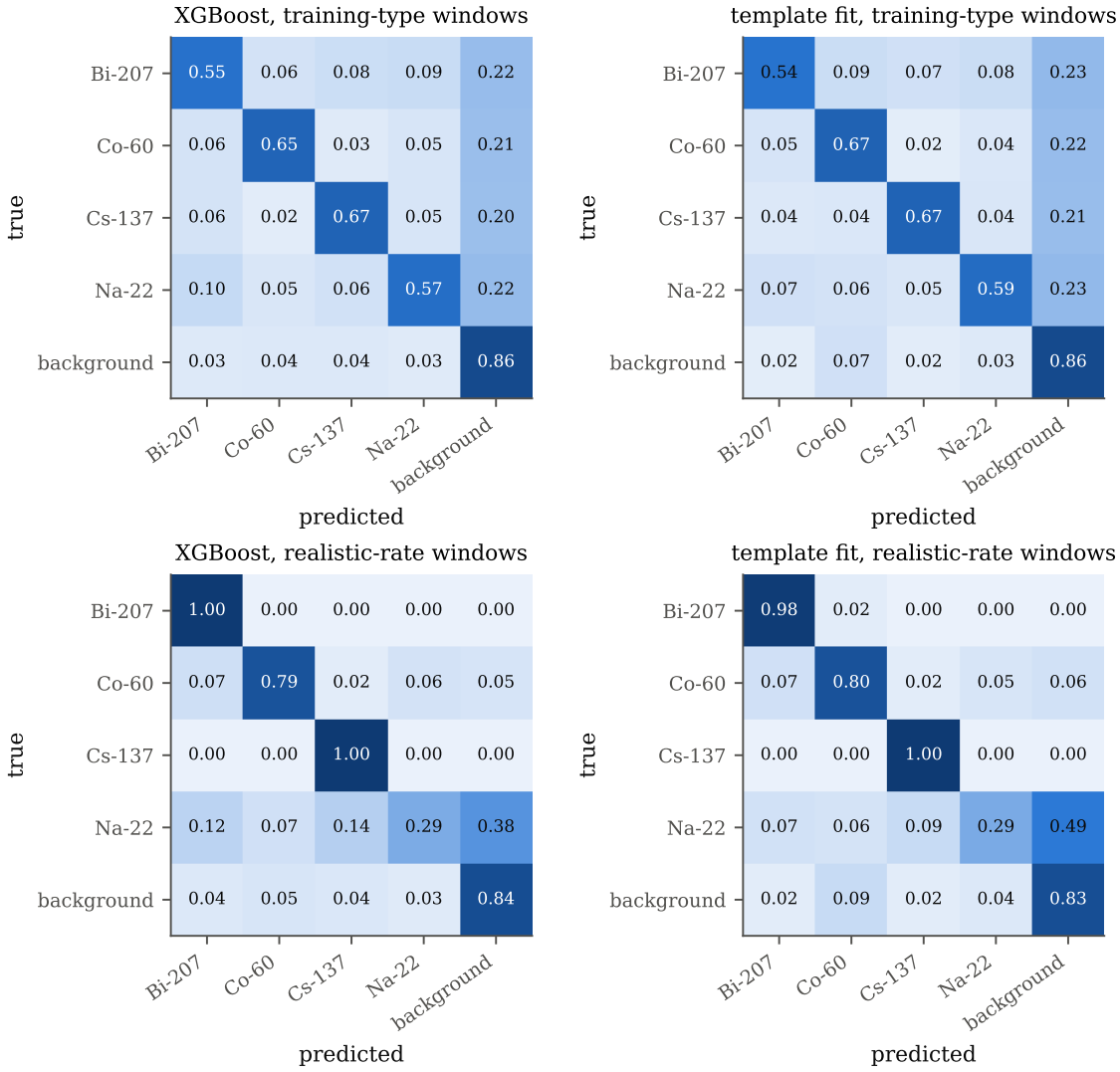


Figure 8: Confusion matrices on the test windows, normalised by true class.

3.6 Co-60 against Bi-207

Co-60 and Bi-207 have gamma lines at similar energies, and we looked at them separately. We define the swap rate as the fraction of Co-60 and Bi-207 windows that are assigned to the other source, among the windows assigned to one of the two. Figure 14 shows it against the number of source events: for windows with at least half of their events from the source, it drops from 24.7% (XGBoost) and 22.0% (template fit) in the lowest band of source events to 0.3% and 2.1% in the band from 300 to 3000, and to 0.1% and 1.0% above 3000. Figure 15 shows that, once the counts are high, the swap rate stays low for all values of the resolution and the energy scale.

This is not the hardest pair of the set. In the confusion matrices of Figure 8, Bi-207 and Na-22 are mistaken for each other more often than Co-60 and Bi-207. The Compton continua of Bi-207 (edges at 0.39 and 0.86 MeV) and Na-22 (0.34 and 1.06 MeV) overlap over most of the range.

3.7 XGBoost against the template fit

Overall, the two classifiers give similar results on both sets of test windows (Figure 9, Table 7). The template fit does better at the lowest numbers of source events, where a model with a small number of free parameters is an advantage. XGBoost does better when the window holds

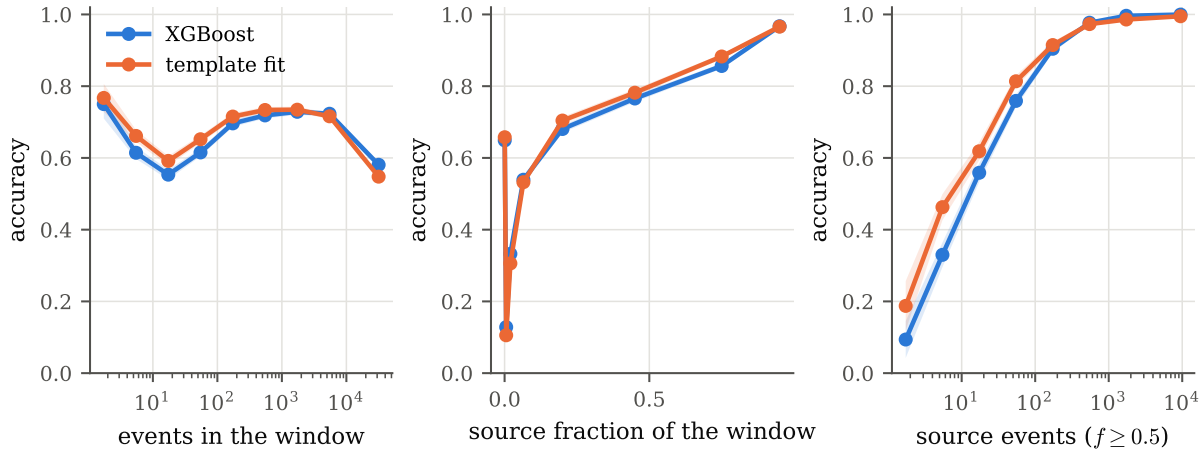


Figure 9: Accuracy on the training-type test windows against the total number of events (left), the source fraction (middle) and the number of source events for windows with a source fraction $f \geq 0.5$ (right). Bands show one standard error.

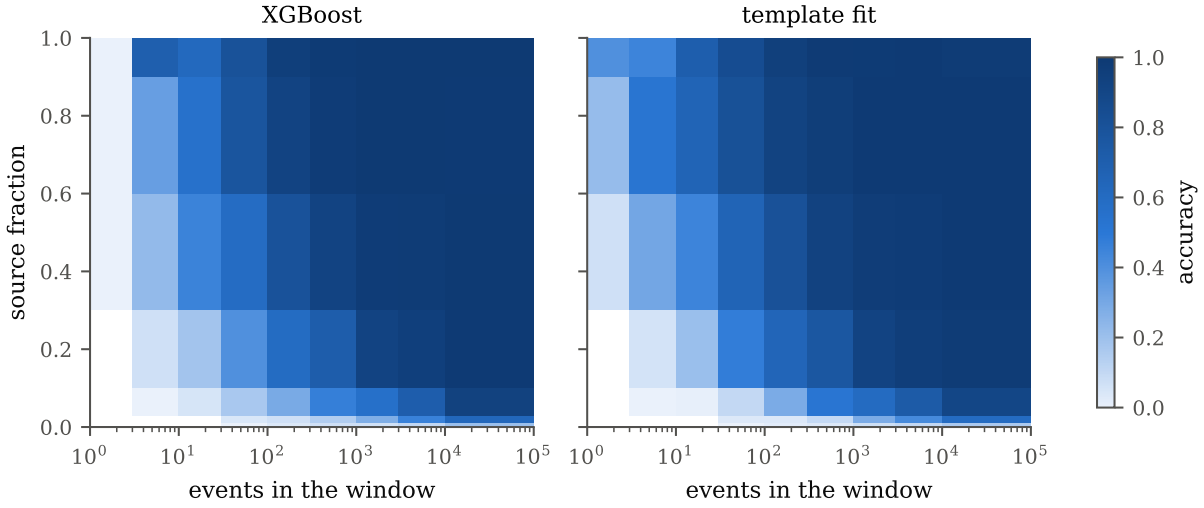


Figure 10: Accuracy as a function of the total number of events and the source fraction.

many source events, where it can use the fine structure of the spectrum, and it makes fewer Co-60/Bi-207 swaps there.

The probabilities of the XGBoost model are well calibrated (Figure 16): the expected calibration error is 0.015 on the training-type windows and 0.043 on the realistic windows (Brier scores 0.424 and 0.300). A confidence value can therefore be used as such.

3.8 Realistic source strengths

The training-type windows scan the number of source events. The realistic windows show what happens with the sources as they are now. Their present activities (Table 1) are about 1.1×10^6 Bq for Cs-137, 1.3×10^5 Bq for Bi-207, 1.2×10^2 Bq for Co-60 and 8.8×10^0 Bq for Na-22. Figure 17 shows the recall for each source and bar, and Figure 18 shows it against the duration of the window.

Cs-137 and Bi-207 are identified in every bar, even in the shortest windows, because they produce more source events than the muons. Co-60 is identified in the bars close to the source, and in the other bars it needs long windows. Na-22, with its very low activity, is recovered in the nearest

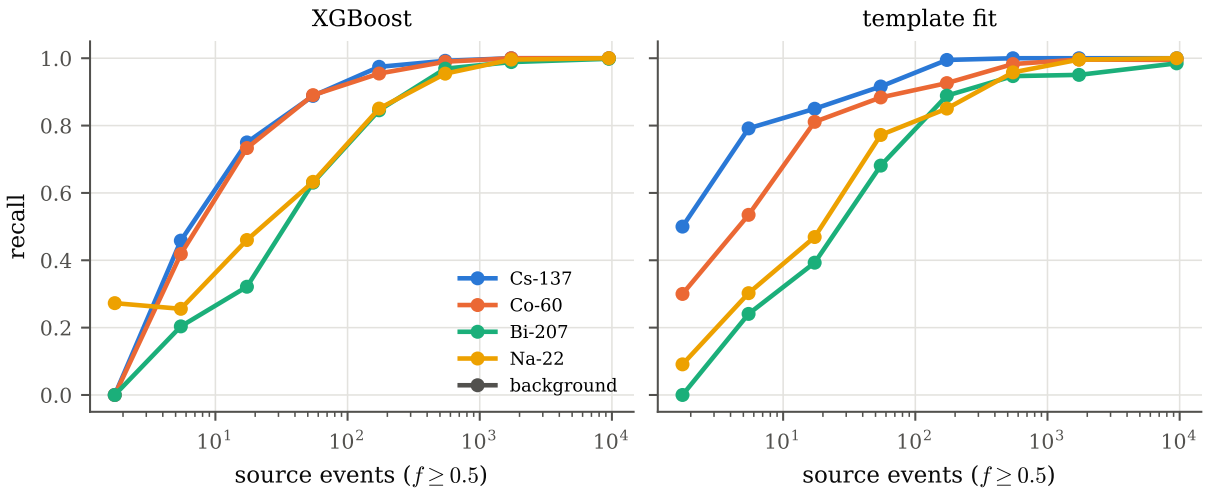


Figure 11: Recall of each class against the number of source events, for windows with a source fraction of at least 0.5.

bars and only partly in the others, even for windows of an hour. The recall of all sources and bars follows the median number of source events per window (Figure 19): a window with around 1000 source events is identified almost always, one with about 100 gives a recall of about 0.7, and one with 20 or fewer gives a recall of the order of 0.1. This is the same relation as in the training-type windows, and the difference between the sources and the bars comes mostly from the number of source events that reach the bar. The position of the capsule along the bars changes the rate in each bar, which the realistic windows include.

3.9 A source that was not in the training

A registry also needs to notice a source it does not know. To test this we trained both methods without Bi-207 and applied them to the windows of that source. Without a dedicated class, these windows are assigned to the nearest known classes (Figure 20): most of them go to Na-22, which has the most similar spectrum, and a large fraction go to background.

We then asked whether a score can separate the unseen source from the known classes. The scores are $1 - p_{\max}$ for XGBoost, where $p_{\max}$ is the highest class probability, and the Poisson deviance per degree of freedom for the template fit. On all windows, the separation is poor (AUC of 0.57 and 0.56), since most windows do not have enough source events to be told from any source (Figure 21, left). On windows with at least 1000 source events and a source fraction of at least 0.9 the score $1 - p_{\max}$ separates the two groups well (Figure 21, right, and Figure 22): the AUC is 0.99 for 1000 to 10 000 source events and 1.00 above 10 000 (Figure 23). The goodness of fit of the template fit is much less effective (AUC 0.67 in the first of these bins). Table 8 gives the fraction of unseen windows flagged at fixed rates of known windows wrongly flagged. No threshold is chosen here: the curves are meant for choosing one once the lab data show how the scores behave.

The probabilities of the known classes in this regime are very close to 1, so the separation relies on the far tail of the score, $1 - p_{\max}$ of order 10^{-4} (Figure 21, right). Such small values will depend on the exact calibration of the model, and a threshold chosen on simulation may not transfer to data.

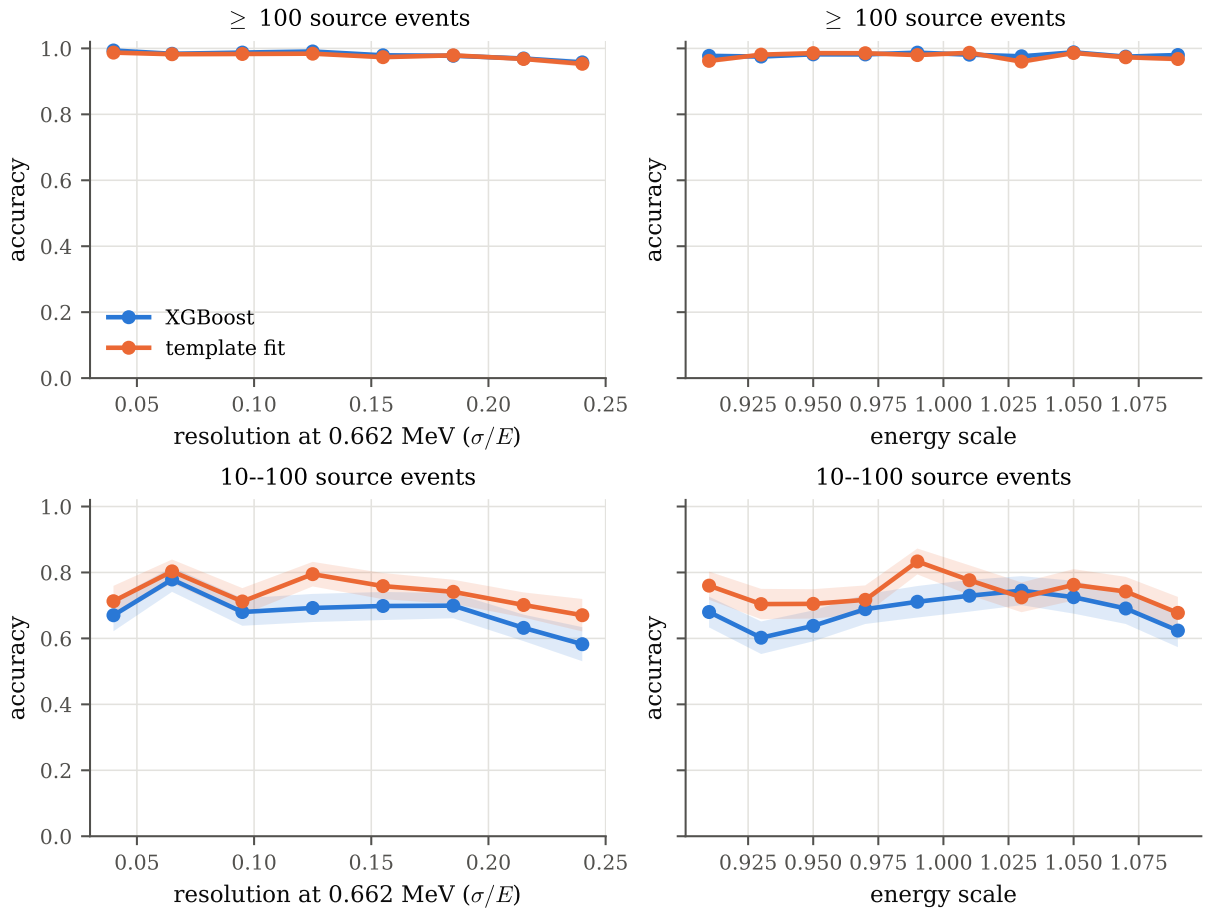


Figure 12: Accuracy against the resolution (left) and the energy scale (right) for windows with a source fraction of at least 0.5, with at least 100 source events (top) and 10 to 100 source events (bottom).

4 Conclusion

We built a chain that goes from the simulation of the lab sources and of cosmic muons to a classifier that names the source seen by a single bar. The result is mostly set by one quantity, the number of source events in the window. With a few hundred source events the identification is correct in most cases, with a few thousand it is almost always correct, and it holds over wide ranges of energy resolution and energy scale. XGBoost and a template fit give similar results; XGBoost has the advantage at high counts, and its probabilities are well calibrated. With the source strengths of today, Cs-137 and Bi-207 are identified in any bar in seconds, Co-60 needs a long window in the bars far from the source, and Na-22 is out of reach except close to the source or with very long windows. A source outside the training set can be flagged when the window has a large number of source events, and not otherwise.

The method is ready to be tested in the lab. The measurements we need are windows of data from each of the four sources, at known positions along known bars, with the duration and the event counts recorded. With them we can:

1. compare the measured spectra with the simulated ones (the positions of the Compton edges, and the shape of the continuum) and the measured event rates with the predicted ones;

2. fit the energy resolution and the energy scale of each bar and replace the nuisance ranges by the measured ones;

3. run the trained model on the measured windows and compare the accuracy with the curves

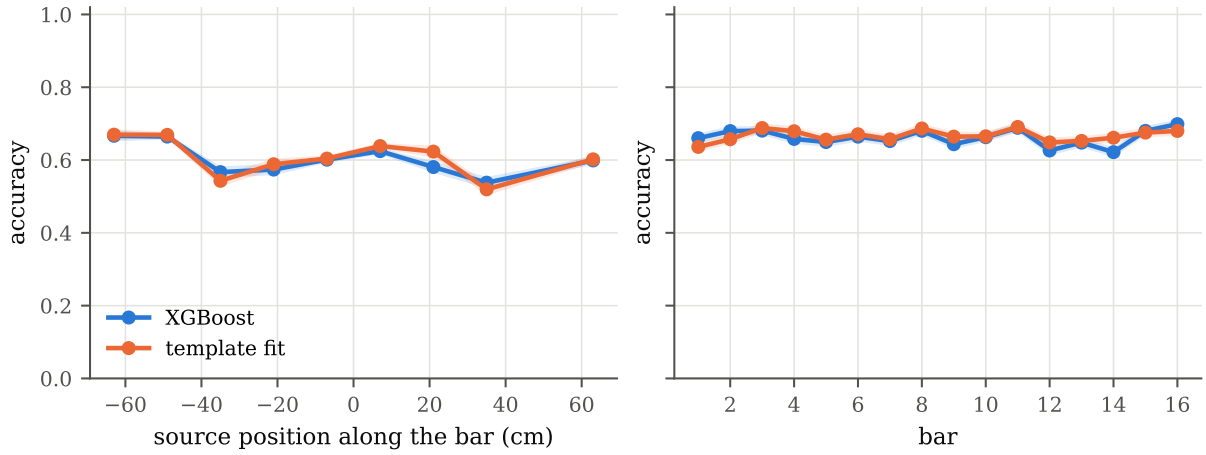


Figure 13: Accuracy against the source position along the bar (left) and the bar (right), on the training-type test windows.

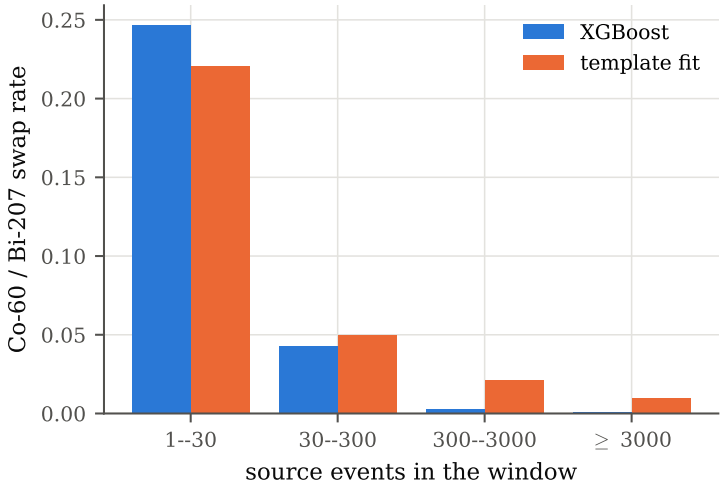


Figure 14: Co-60/Bi-207 swap rate for windows with a source fraction of at least 0.5, in bands of source events.

of this report, in particular the dependence on the number of source events;

4. choose the thresholds for "too few events to decide" and for "unknown source" from the measured score distributions, since the simulation alone cannot fix them.

Retraining with the measured response is cheap, so the comparison with the lab data can be repeated as often as needed.

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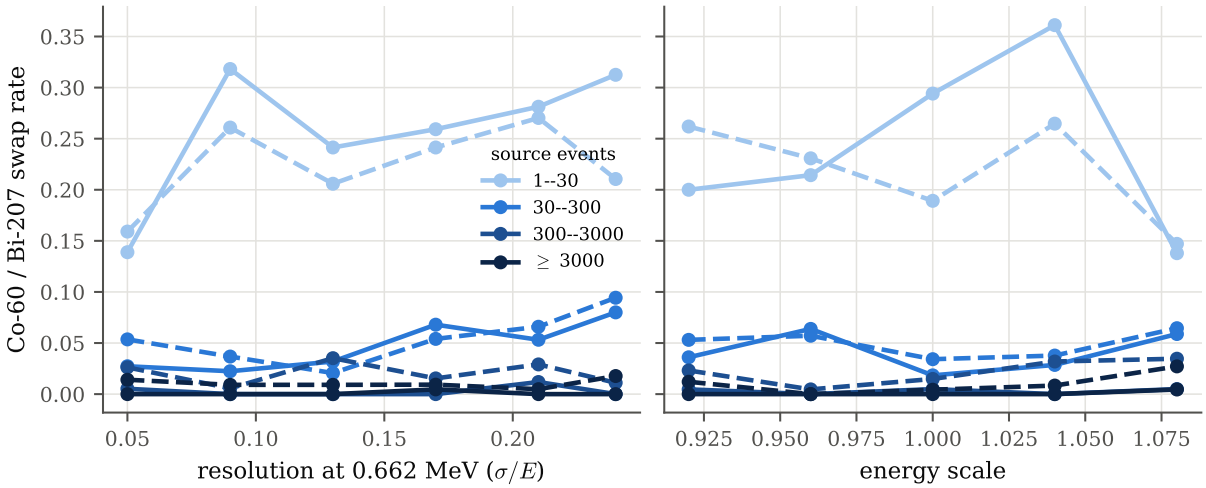


Figure 15: Co-60/Bi-207 swap rate against the resolution (left) and the energy scale (right), in bands of source events. Solid: XGBoost. Dashed: template fit.

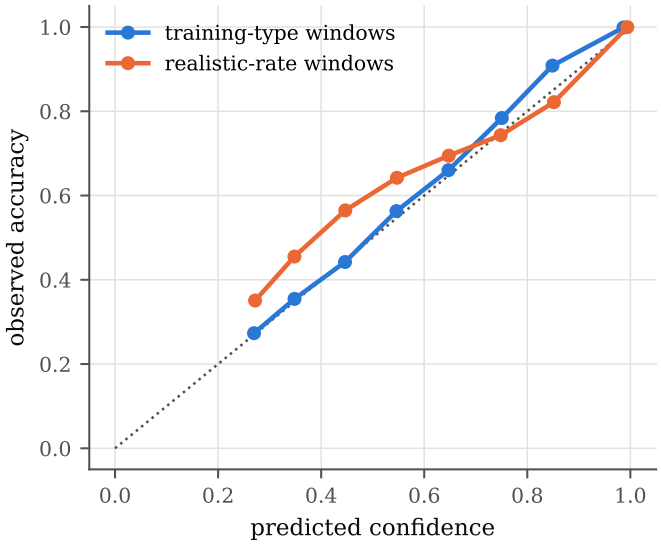


Figure 16: Reliability diagram of the XGBoost top-class probability.

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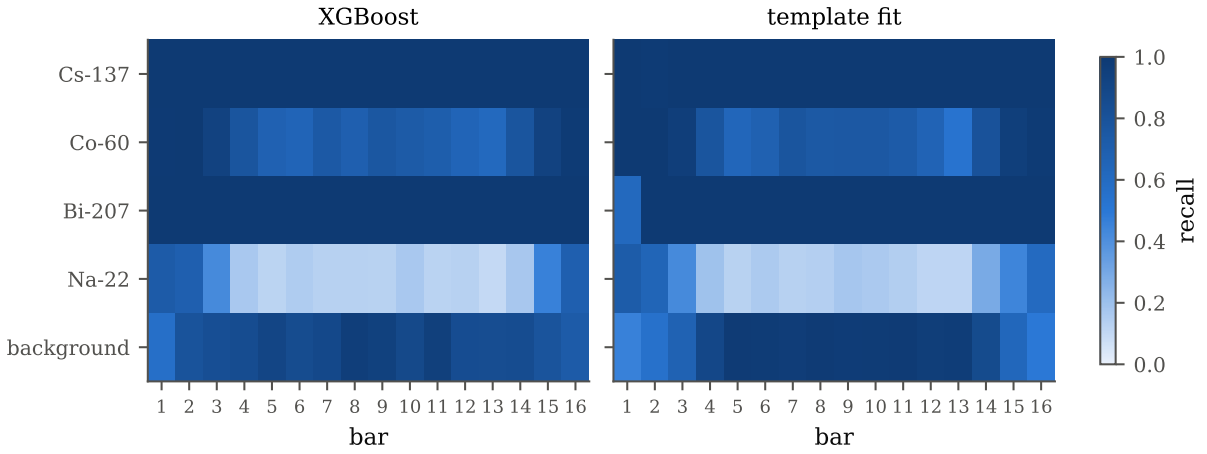


Figure 17: Recall for each source and bar on the realistic windows (1 to 3600 s).

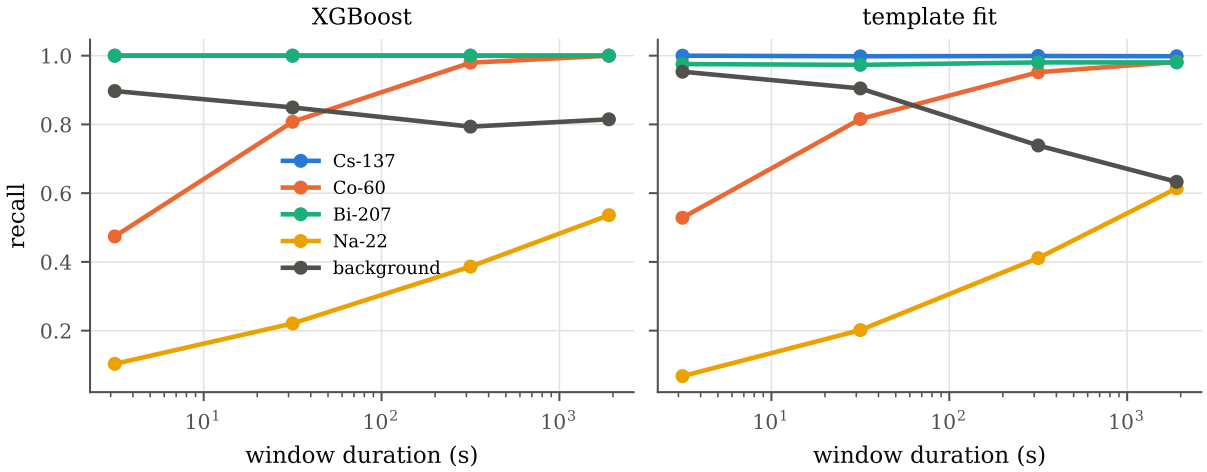


Figure 18: Recall against the window duration on the realistic windows.

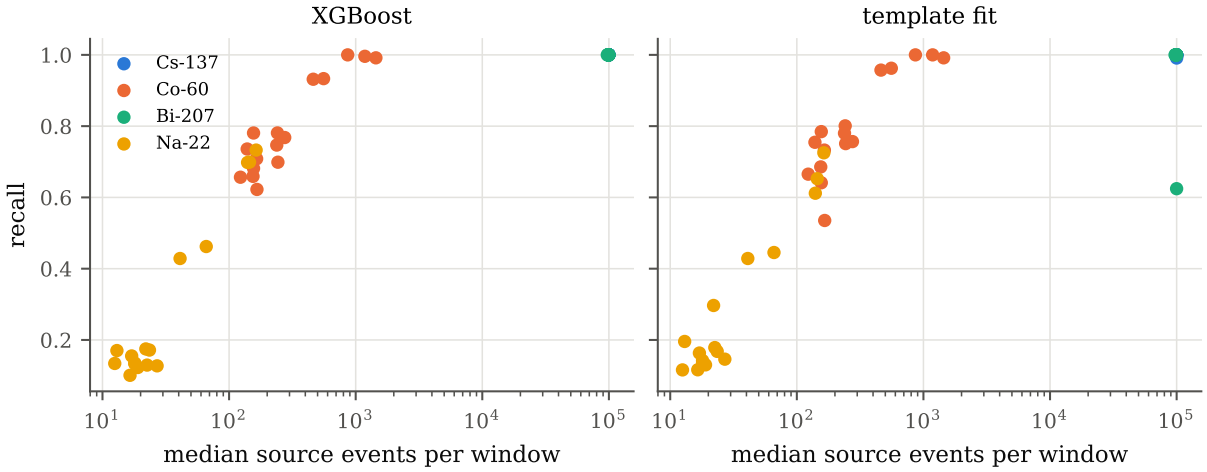


Figure 19: Recall against the median number of source events per window. Each point is one source in one bar.

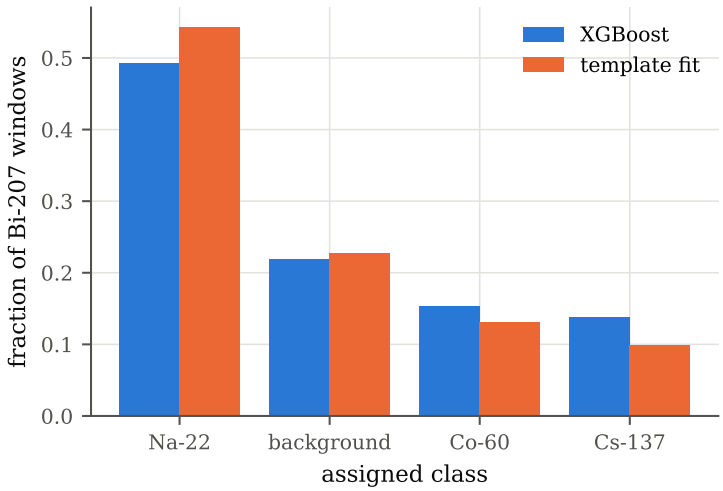


Figure 20: Classes assigned to the windows of the left-out source.

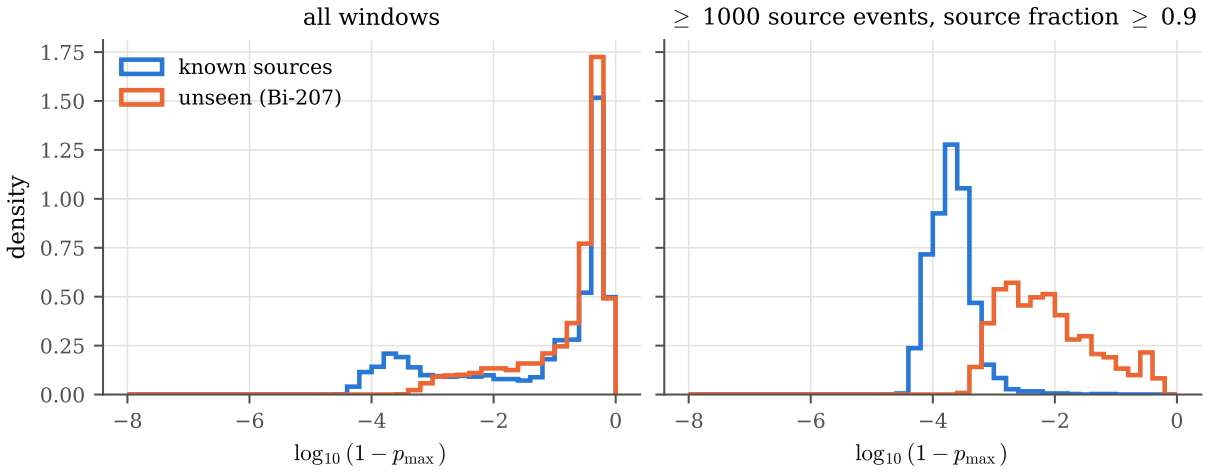


Figure 21: Distribution of $\log_{10}(1 - p_{\max})$ for the known classes and for the left-out source. Left: all windows. Right: windows with at least 1000 source events and a source fraction of at least 0.9.

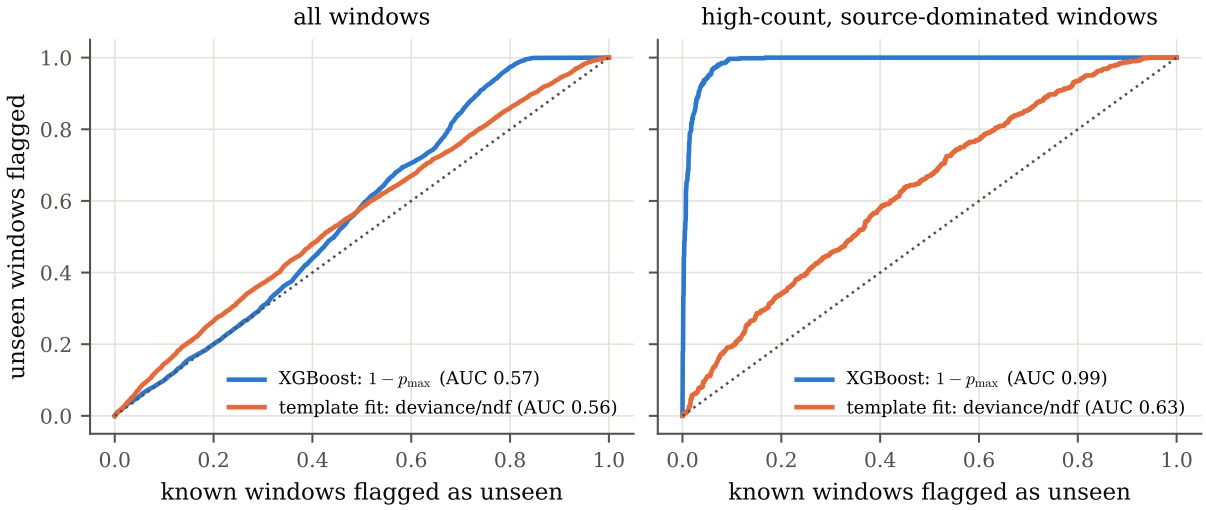


Figure 22: ROC curves for flagging the left-out source, for all windows (left) and for high-count, source-dominated windows (right).

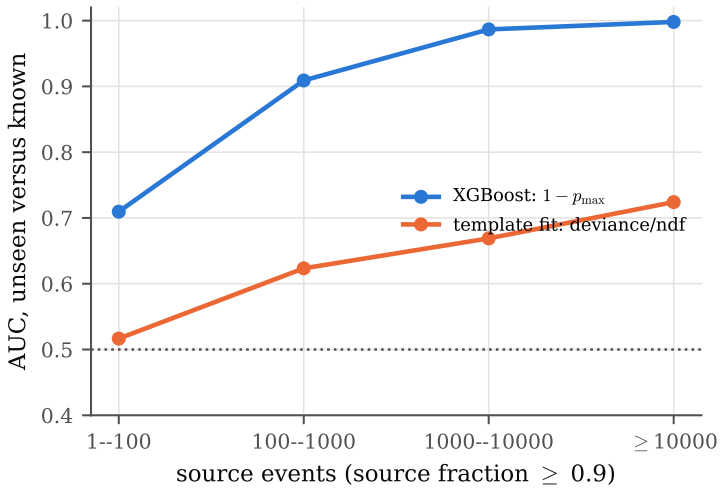


Figure 23: AUC for separating the left-out source from the known classes, against the number of source events, for source-dominated windows.

window selection	TPR (%) at FPR 1%	TPR (%) at FPR 5%	TPR (%) at FPR 10%
all windows	2	5	10
100–1000 source events	7	40	65
1000–10000 source events	67	94	99
$\geq$ 10000 source events	93	99	100

Table 8: Percentage of left-out windows flagged by the score $1 - p_{\max}$ at three rates of known windows flagged (FPR), for several selections of windows.